Class #11 - Orthogonal Sets

1. Def: A set of vectors $\{\vec{u}_1, \cdots, \vec{u}_p\}$ in $\mathbf{R^n}$ is said to be an *orthogonal set* if each pair of distinct vectors from the set is orthogonal, i.e., $\vec{u}_i \cdot \vec{u}_j = 0$ whenever $i \neq j$.

ex $\mathbb{R}^2$

$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ ✗

$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix} \right\}$ ✓

$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ L.I.

Not orthogonal set.

$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$ Not L.I.

orthogonal set

2. Theorem: If $S = \{u_1, \cdots, u_p\}$ is an orthogonal set of nonzero vectors in $\mathbf{R^n}$, then S is linearly independent.

(Note: If $p = n$, S is a basis for $\mathbf{R^n}$.)

Proof:

Have: $\{\vec{u}_1, \cdots, \vec{u}_p\}$ in $\mathbb{R}^n$ $\quad \vec{u}_i \neq 0 \ \forall i$

$\vec{u}_i \cdot \vec{u}_j = 0 \quad \forall i \neq j$

To show: $\{\vec{u}_1, \cdots, \vec{u}_p\}$ L.I.

PF Suppose $\{\vec{u}_1, \cdots, \vec{u}_p\}$ L.D.

There exists $\vec{u}_i$ s.t. $c_1\vec{u}_1 + \cdots + c_{i-1}\vec{u}_{i-1} = \vec{u}_i$

$\vec{u}_i \cdot (\quad) = \vec{u}_i \cdot \vec{u}_i$

$\Rightarrow \sum_{k=1}^{i-1} c_k (\vec{u}_i \cdot \vec{u}_k) = \|\vec{u}_i\|^2$

$\vec{u}_i \cdot \vec{u}_k = 0 \quad \forall k = 1, \cdots, i-1$

$\Rightarrow \|\vec{u}_i\|^2 = 0 \iff \vec{u}_i = 0$

But $\vec{u}_i \neq 0$. This is a contradiction!

3. Def: An *orthogonal basis* for a subspace W of $\mathbf{R^n}$ is a basis for W that is also an orthogonal set.

4. Theorem: Let $\{u_1, \cdots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbf{R^n}$. Then each $y \in W$ has a Unique representation as a linear combination of $u_1, \cdots, u_p$. In fact, if

$$y = c_1u_1 + \cdots + c_pu_p$$

then

$$c_i = \frac{y \cdot u_i}{u_i \cdot u_i} \quad (i = 1, \cdots, p).$$

Proof:

For uniqueness, suppose $y = d_1u_1 + \cdots + d_pu_p$.

So, $0 = (c_1 - d_1)u_1 + \cdots + (c_p - d_p)u_p$

$\{u_1, \cdots, u_p\}$ L.I. $\Rightarrow c_i - d_i = 0 \ \forall i$

$\Rightarrow c_i = d_i \ \forall i$

Now, to find c_i

$\vec{u_i} \cdot \vec{y} = \vec{u_i} \cdot (c_1\vec{u}_1 + \cdots + c_p\vec{u}_p)$

$\vec{u_i} \cdot \vec{y} \Rightarrow c_i(\vec{u_i} \cdot \vec{u_i})$

$\therefore c_i = \dfrac{\vec{y} \cdot \vec{u_i}}{\vec{u_i} \cdot \vec{u_i}}$

$\{b_1, \cdots, b_n\}$ basis of $\mathbb{R}^n$

$x \in \mathbb{R}^n$

$x = c_1b_1 + \cdots + c_nb_n$

To find c_i.

$$\left[\begin{array}{ccc|c} b_1 & \cdots & b_n & \begin{matrix} x_1 \\ \vdots \\ x_n \end{matrix} \end{array}\right] \sim \left[\begin{array}{c|c} & \end{array}\right]$$

5. Example: Let $S = \{u_1, u_2, u_3\}$ where

$$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}.$$

Can Check $u_i \cdot u_j = 0 \ \forall i \neq j$

(a) S is an orthogonal basis of $\mathbb{R}^3$.

(b) If $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$, find $[y]_S = \begin{bmatrix} \frac{y \cdot u_1}{u_1 \cdot u_1} \\ \frac{y \cdot u_2}{u_2 \cdot u_2} \\ \frac{y \cdot u_3}{u_3 \cdot u_3} \end{bmatrix}$

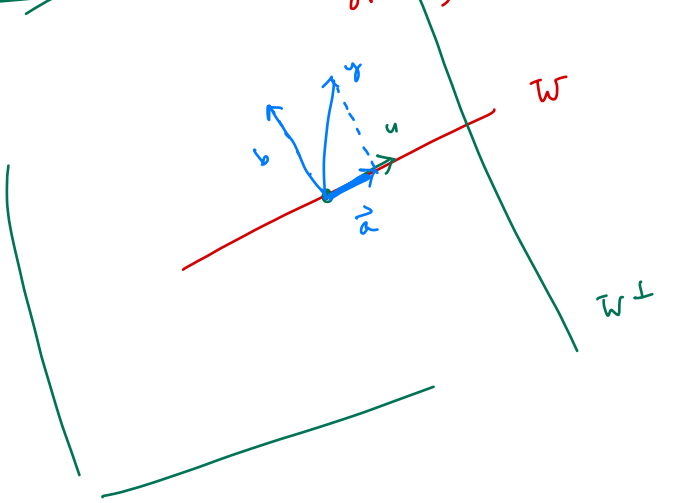
$y = c_1u_1 + c_2u_2 + c_3u_3$

$[y]_S = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$

$$= \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} | & | & | & \\ u_1 & u_2 & u_3 & \begin{matrix} 6 \\ 1 \\ -8 \end{matrix} \\ | & | & | & \end{array}\right] \sim$$

6. Fix a nonzero vector $u \in \mathbf{R^n}$. (Then $W = \text{Span}\{u\}$ is a 1-dimensional subspace of $\mathbf{R^n}$.) Given a vector $y \in \mathbf{R^n}$, decompose y into sum of two vectors a and b so that $a \in W$ and $b \in W^\perp$ (that is, a and b are orthogonal).



$y = a + b$ (a ∈ W, b ⊥ u)

$u \cdot y = u \cdot (cu + b)$

$u \cdot y = c(u \cdot u) + u \cdot b$ ($u \cdot b = 0$ since $u \in W$, $b \in W^\perp$)

$c = ?$

$\therefore c = \dfrac{u \cdot y}{u \cdot u}$

$a = cu = \left(\dfrac{y \cdot u}{u \cdot u}\right) u$ is the *orthogonal projection of y onto* W (or *onto* u), and

$b = y - a$ is the *orthogonal projection of y onto* $W^\perp$ (b is called the *component of y orthogonal to u*).

7. Example: Let $y = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Write y as the sum of two orthogonal vectors, one in $\text{Span}\{u\}$ and one orthogonal to u.

$$a = cu = \left(\frac{\begin{bmatrix} 7 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix}}{\begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix}}\right) \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \frac{28+12}{16+4} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

$$b = y - a = \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$y = \begin{bmatrix} 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$\begin{bmatrix} 8 \\ 4 \end{bmatrix}$ in $\text{Span}\left\{\begin{bmatrix} 4 \\ 2 \end{bmatrix}\right\}$ (1)

8. Def: A set $\{u_1, \cdots, u_p\}$ is an *orthonormal set* if it is an orthogonal set of unit vectors.
Note: If $W = \text{Span}\{u_1, \cdots, u_p\}$, then $\{u_1, \cdots, u_p\}$ is an orthonormal basis of the subspace W.

$\|u_i\| = 1 \quad \forall i$
$\Leftrightarrow \|u_i\|^2 = 1$
$\Leftrightarrow u_i \cdot u_i = 1$

L.I.

$u_i \neq 0 \quad \forall i$

$y \in W$

$y = c_1 u_1 + \cdots + c_p u_p$

$c_i = y \cdot u_i \quad \forall i$

9. Example: Generate at least two distinct orthonormal bases for $\mathbf{R^3}$.

plural

$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$

$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{matrix} \\ u_3 \end{matrix}$

$W = \text{Row } A$

$W^\perp = \text{Nul } A$

Find u_3 in Nul A

$W = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

orthonormal basis of $\mathbb{R}^3$ $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\|\cdot\|} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\|u_3\|} u_3 \right\}$

① Attendance

② Per. quiz

③ Class # 11 ~ #8

④ Quiz #5 Avg. 6.55

⑤ Midterm: Monday Class #1 ~ today
PP #1 ~ #15

D: F MS: T
DG: F MG: T
DM: F
DY: F

① T/F Sometimes, an orthogonal set is L.D. (F)

② T/F Orthogonal set of nonzero vectors is always L.I. (T)

③ T/F If $\{u_1, \dots, u_n\}$ orthogonal basis of $\mathbb{R}^n$, and $x \in \mathbb{R}^n$, then x can be written $c_1u_1 + \dots + c_nu_n$, and also $d_1u_1 + \dots + d_nu_n$ (F)
where $c_j \neq d_j$ for some j.
J: F
CH: F

④ T/F W : subspace of $\mathbb{R}^n$ $\{u_1, u_2, \dots, u_p\}$ an orthogonal basis of W.
$u_i \cdot u_i = 1$
$\|u_i\|^2 = 1$
$\in W$
$y = c_1u_1 + \dots + c_pu_p$
Then, $c_i = \frac{y \cdot u_i}{u_i \cdot u_i}$ $(i = 1, \dots, p)$
T: T
J: F
Y: F

⑤ $\mathbb{R}^n$
$W = \text{span}\{u\}$

Take y in $\mathbb{R}^n$
$y = a + b$
$a \in W$ $b \in W^\perp$

Name:
$a =$ the orthogonal projection of y onto W $= \frac{y \cdot u}{u \cdot u}u$